

IntelliCV Relational Mapping

Relational Mapping

USER

<u>user_id</u>	name (f, m, l)	email	registration_date	profile_summary	phone
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DOCUMENT

<u>doc_id</u>	<i>user_id</i>	file_name	base_name	ext	text_extracted	upload_date	keywords
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PROJECT

<u>proj_id</u>	<i>user_id</i>	title	description	start_date	end_date	project_length	tech_stack
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CERTIFICATE

<u>cert_id</u>	<i>user_id</i>	title	issuing_org	issue_date	validity_period
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JOB_DESCRIPTION

<u>job_id</u>	user_id	title	company_name	location
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GENERATED_RESUME

<u>resume_id</u>	user_id	job_id	match_score	generated_text	timestamp
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1. First Normal Form (1NF)

1NF Violations (Multivalued Attributes)

The ER Diagram contains Composite Attributes (e.g., Name) and Multivalued Attributes (e.g., Phone, Tech Stack, Keywords, Skills). In a direct relational mapping, these would violate 1NF if kept in a single table.

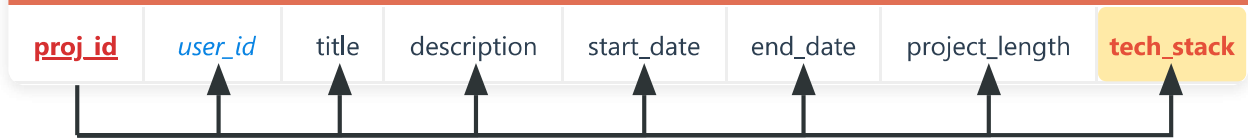
USER (1NF VIOLATION)

<u>user_id</u>	name (f, m, l)	email	registration_date	profile_summary	phone
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DOCUMENT

<u>doc_id</u>	user_id	file_name	base_name	ext	text_extracted	upload_date	keywords
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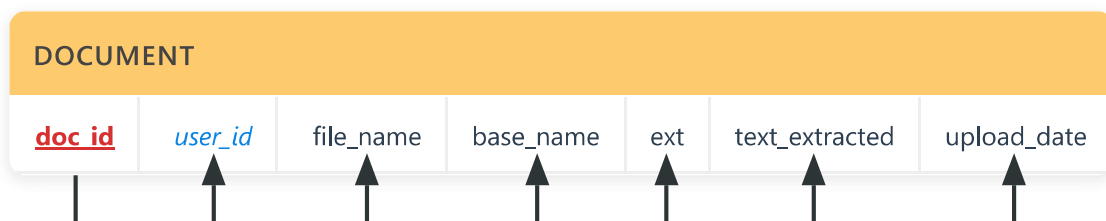
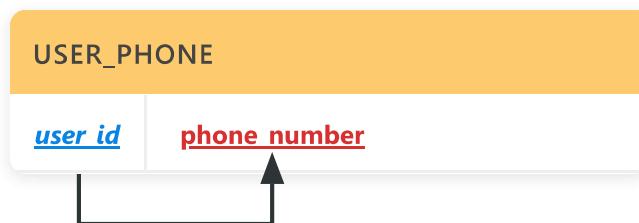
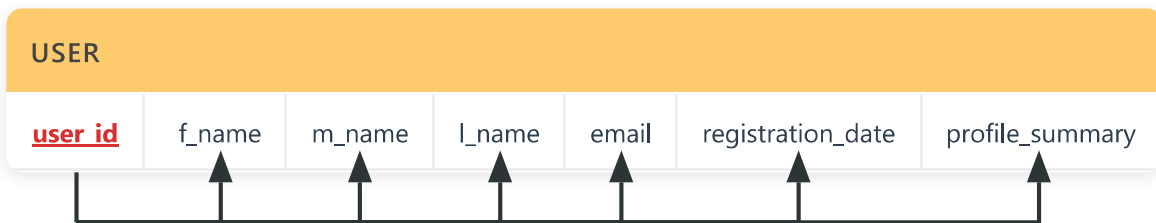
PROJECT (1NF VIOLATION)

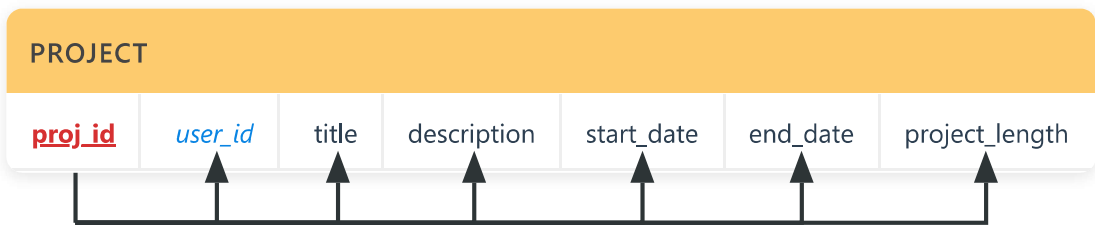


2. Second Normal Form (2NF)

Partial Dependencies Removed /
Multivalued Fixed

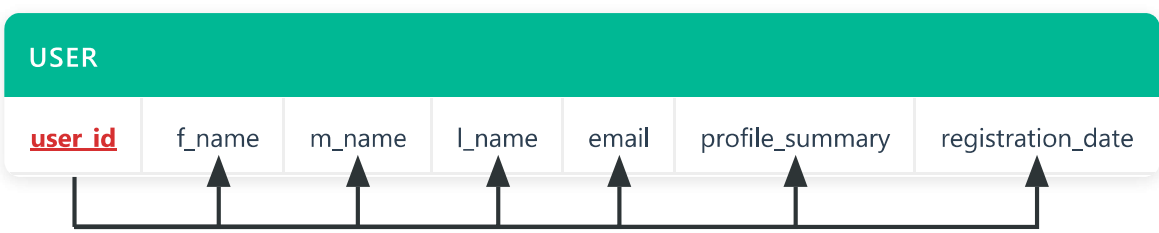
Multivalued attributes are now in separate tables (`User\_Phone`, `Project\_Tech\_Stack`).
Primary keys are atomic.

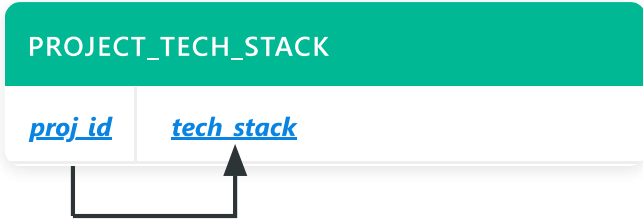
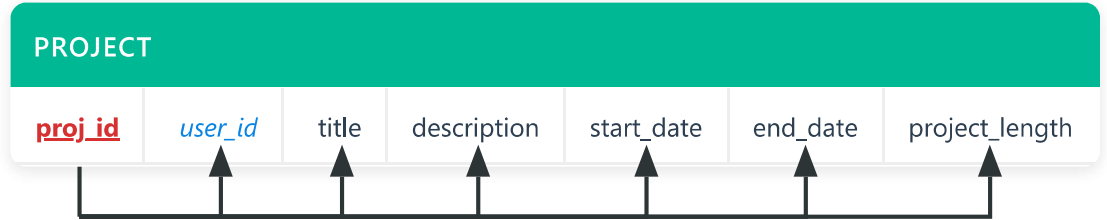
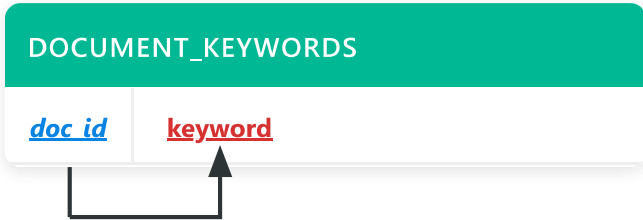
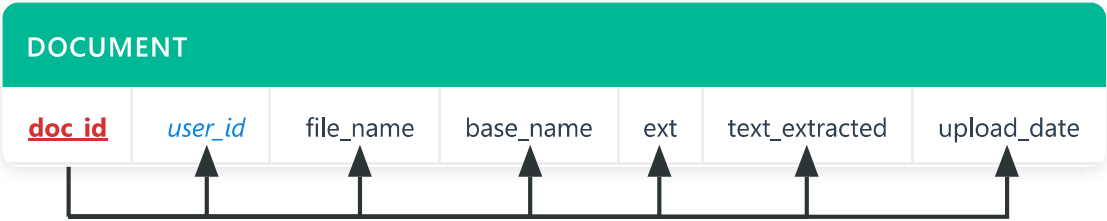
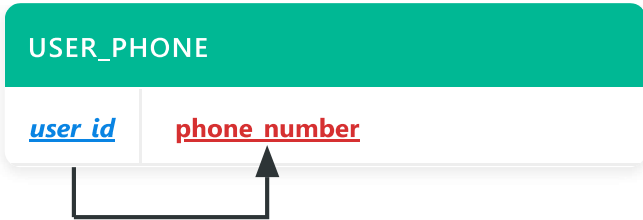




3.Third Normal Form (3NF)

Normalized Structure





CERTIFICATE

<u>cert_id</u>	user_id	title	issuing_org	issue_date	validity_period
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JOB_DESCRIPTION

<u>job_id</u>	user_id	title	company_name	location
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JOB_REQUIRED_SKILLS

<u>job_id</u>	<u>skill</u>
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GENERATED_RESUME

<u>resume_id</u>	user_id	job_id	match_score	generated_text	timestamp
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